AN ECLECTIC TECHNOLOGIES DETERMINATION UNDER FD-BL

FD-BL-D2 — Indication Integrity: Trustworthy Basis for Authorization

Determination under FD-BL

Version 0.5 — DRAFT · 15 October 2026

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Designation: FD-BL-D2
Class: Determination under FD-BL (not a Foundational Definition — see §2)
Governing instrument: FD-BL — Foundational Definition — The Bright Line. This instrument determines the question FD-BL carried at §6.2, which FD-BL now closes to a pointer here. Versions are held in FD-REG.
Version: 0.5 — DRAFT (issued in the November 2026 Edition; supersedes v0.4)
Date: 15 October 2026
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Organization: Eclectic Technologies
Companion determination: FD-BL-D1 (mode of discharge); this instrument settles the indication-integrity floor D1 §9 held on an interim basis
Companion constructions: FD-BL §5 situational-awareness example; DBA-W §5.5.4 (operator-facing decision support); DBA-GAS-FC2 (systematic-encroachment detection)

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1. Purpose

FD-BL §4.5 requires that an action placed above the Bright Line have substantive, attributable human authorization by a qualified Command Broker, enforced architecturally. FD-BL-D1 permits, for a narrow class, that authorization to be discharged in advance by the Command Broker, with a verified gatekeeper admitting each action. Both modes share one dependency, unstated before this instrument:

the authorization is sound only if the **state it is granted against is real**. FD-BL §6.2, as it stood at v0.2, left open whether a below-line placement is available to generative output that functions as the basis for such an authorization, and recorded a recommended disposition. This instrument determines that question.

This determination resolved FD-BL §6.2. Since FD-BL v0.3, §6.2 has been a pointer to this instrument, which carries the resolution and its amendment set. It also **settles the interim floor** stated at FD-BL-D1 §9: the requirement that a gatekeeper's inputs be traceable to independently verifiable values ceases to be interim and becomes the general rule stated here, binding on the human-authorization case as well.

2. Authority, class, and what this does not alter

2.1 Class

FD-BL-D2 is a **determination**, not a foundational definition. By the family test at FD-BL §2.2, a concept belongs to the Foundational Definitions family only when the correctness of other documents depends on agreeing what *kind of thing* it is. This instrument does not define a kind of thing; it attaches an **obligation** to output already placed by the definition. It is therefore designated as a determination bound to FD-BL rather than as a peer of FD-BL or FD-LD, and it is a sibling of FD-BL-D1.

2.2 What this determination may and may not do

This determination operates under the constraint FD-BL §6.2 placed on it at v0.2, which FD-BL §6 records as honored and which is restated here as binding:

It attaches a qualifier to output of an in-scope component that can function as the basis for an above-line authorization, or as an input on which placement or discharge depends. It does not vary placement (FD-BL §4.2), granularity (FD-BL §4.3), or scope (FD-BL §4.4). Output subject to this qualifier remains placed where its own consequence puts it — including below the line where that is where consequence puts it. Nothing here reclassifies indication, synthesis, or advisory output to above the line, and nothing here creates an intermediate class. The remedy is an obligation that rides on certain output, not a movement of the line — exactly as FD-BL §6.2 directs.

2.3 The governing principle it applies

FD-BL §4.2 places output that would make no change of its own below the line as a process action, and provides that where it can function as the basis for an above-line authorization, that role attracts the obligation of this instrument rather than moving it across the line. The principle this determination applies follows directly: an authorization — whether the in-the-moment judgment of a Command Broker (FD-BL §4.5) or the advance proof accepted for a gatekeeper (FD-BL-D1) — is **bounded only if the state it is granted against is trustworthy**. A broker confirming an above-line action against a corrupted or drifted synthesis is confident and wrong; a gatekeeper evaluating a proof against corrupted indication has proven a proposition about a fiction. In both cases the authorization the framework relies on is hollowed out with no protective layer registering a fault. This determination makes the trustworthiness of that basis a named precondition of the authorization, rather than an unstated assumption.

3. The criterion: an authorization is bounded only if its basis is trustworthy

Whether the basis of an above-line authorization is trustworthy decomposes into two parts, and the criterion applies to **both**:

- **Provenance** — can the output be traced to values confirmable independently of the generative system that produced it?
- **Singularity** — is that output the *only* thing standing between the authorizer and the action, or is there an independent path by which the state can be confirmed?

Output that is traceable but sole still fails when the traceable channel is itself the one that drifted; output that is corroborated but untraceable still fails when neither channel is anchored to anything real. Both parts are required. The requirements in §4 are the conditions under which both hold.

4. The determination

Output described in I1 carries the indication-integrity obligation, whatever its own placement. The obligation is discharged only where I2 and I3 both hold. Where they do not, that output may not serve as the basis for an above-line authorization, nor as an input on which placement or discharge depends.

4.1 I1 — The obligation attaches by function, not by object type

The qualifier attaches to any output of an in-scope component (FD-LD §4.3) that **can function as the basis for an above-line authorization** — a Command Broker's confirmation under FD-BL §4.5, or a verified gatekeeper's deterministic check under FD-BL-D1 — **or as an input on which placement or discharge depends**, such as the state the consequence classifier routes on (FD-BL §4.3; FD-BL-D1 §5). It attaches by that function alone, regardless of whether the output is a display, a synthesis, a summary, or a recommendation, and regardless of the output's own below-line placement. Output that can function as neither — output with no authorizing role and no role in placement or discharge — does not carry the obligation. The trigger is the authorizing function fixed by consequence (§2.3), not the artifact's form and not a reclassification of the artifact. Where this instrument speaks of a generative system, a generative synthesis or a generative recommendation, it names the principal case of an in-scope producer, and every requirement it states of them applies to the output of any in-scope component.

4.2 I2 — Traceability to independently verifiable values

Output subject to I1 shall be **traceable to independently verifiable values**. A value is *independently verifiable* when it can be confirmed through a path independent of the generative system that produced the output — a direct read of the underlying measurement, a diverse or redundant instrument, or a deterministic sensor channel — such that a fault, drift, or manipulation in the generative synthesis does not also corrupt the confirming path. A synthesis that the generative system both produces and certifies is not independently verifiable. The determination fixes this principle; the **grade** — degree of redundancy, sensor diversity, cross-check cadence, tolerance — is set at the sector annex (§7).

4.3 I3 — Not the sole basis (independent corroboration required)

Output subject to I1 **shall not be the sole basis** for an above-line authorization. There shall be an independent path by which the authorizer — human Command Broker or verified gatekeeper — can confirm the state before the authorization is valid; the generative synthesis alone shall not carry an above-line authorization across the line. **Where no independent confirming path exists, the output cannot serve as the basis for an above-line authorization at all**, and the authorization is unavailable

until such a path is supplied. This is a hard requirement, not a default; a sector annex may set its grade (§7) but may not waive corroboration for an above-line authorization.

Classifier inputs. Where output subject to I1 serves as an input on which placement depends and cannot be corroborated through an independent path, the action it bears on shall not proceed without in-the-moment Command Broker authorization. This rule governs discharge, not placement: it does not place the action above the line, it changes no placement, and it creates no class of action.

Sufficiency. An indication can be correct and insufficient. Correctness is necessary and not sufficient: where the basis for an authorization is correct but does not show what the authorization depends on, what is missing is identified and sought before the authorization is relied on, and a basis acted on while incomplete is recorded as incomplete. How what is missing is identified, and what may be done while it is sought, is open at this version.

4.4 What changes, and what does not

Under I1–I3 no object moves across the line. What changes is that certain below-line output now carries a precondition: when it grounds an above-line authorization it must be anchored (I2) and corroborated (I3), or the authorization it would ground is not available. Placement, granularity, and scope are untouched (§2.2). The obligation is discharged at the point the authorizer relies on the output — in the moment for a Command Broker, and at proof-acceptance and runtime for a gatekeeper — and it fails safe: absent a trustworthy basis, the above-line action does not proceed on that basis.

5. Relationship to FD-BL-D1 and the two-object model

This determination governs **both** authorization modes with one rule, and unifies the machine and human cases FD-BL-D1 §9 could only partly reach:

- **The human case (FD-BL §4.5).** The Command Broker's confirmation is substantive only against a trustworthy picture. I1–I3 make indication integrity a precondition of the in-the-moment authorization, closing the "confident and wrong" hole named in the FD-BL §5 situational-awareness example.
- **The machine case (FD-BL-D1).** A gatekeeper is bounded by proof only if its inputs are trustworthy (FD-BL-D1 §9). That interim floor is **now settled** by I2–I3: the measured state on which the gatekeeper's deterministic check operates must be independently verifiable and independently corroborated. FD-BL-D1 §9's indication-integrity paragraph is superseded by this instrument; FD-BL-D1 conformed at its v0.2, and its §9 points here.

In the two-object model of FD-BL-D1 §5, **both** objects consume indication: the consequence classifier routes on the state it is shown, and the gatekeeper vetoes on the state it is shown. I1–I3 bind the inputs to both: I1 reaches the classifier's inputs because placement depends on them, and the gatekeeper's because discharge does. This determination does not alter the two-object model; it secures the inputs on which both objects depend.

6. Foreclosures

Two routes are foreclosed and shall not be reopened:

- **"Below the line, therefore unconstrained" is foreclosed.** That a synthesis changes no parameter does not exempt it from the qualifier when it can function as the basis for an above-line

authorization. Below-line placement carries no exemption from I1–I3; this is the precise hole FD-BL §6.2 closes.

- **"Feeds an authorization, therefore above the line" is foreclosed.** Reclassifying all authorization-feeding output as above the line is a placement change FD-BL §6.2 disavows, and it would make role, not consequence, determine placement — the mirror image of the composite-classification and third-tier errors foreclosed under FD-BL §6.1. The remedy is a qualifier that attaches an obligation, never a reclassification that moves the line.

The determination holds the narrow middle: the output stays where consequence puts it, **and** carries the traceability and corroboration obligation whenever it can ground an above-line authorization.

7. Sector instantiation requirements

This determination is sector-neutral. Sector annexes shall instantiate it, and shall in particular specify:

- **The grade of "independently verifiable."** The degree of redundancy, the sensor diversity, the cross-check cadence, and the tolerances that make a value independently verifiable (I2) for that sector's consequence class. The determination requires independence of failure between the generative synthesis and the confirming path; the annex sets how much independence, and at what protection grade. For nuclear, that grade belongs to the nuclear annex.
- **The corroboration path for above-line authorization.** How the independent confirming path required by I3 is realized operationally — a diverse instrument the broker reads directly, a deterministic monitor the gatekeeper cross-checks against, an anomaly detector that flags divergence between the synthesis and the measured trajectory (as DBA-GAS-FC2's systematic-encroachment detection already does for the machine case).
- **Denied-indication behavior.** What the system does when the basis is not trustworthy — when the confirming path is unavailable, or the synthesis and the independent channel diverge beyond tolerance. The above-line authorization shall be unavailable on that basis; re-establishment shall require a deliberate, authenticated act, consistent with the denied-condition immutability FD-BL-D1 §7 requires of the envelope.
- **The operator confirmation step.** The instantiation of I3 in an operator-facing workflow: the confirmation step shall involve independent verification of the generative recommendation, not action on it, and the workflow shall not present generative output as an actionable default or pre-populate control inputs from it (as DBA-W §5.5.4 already requires for water).

8. Accountability

Accountability for an above-line authorization rests with the **named Command Broker who grants it, in the moment or in advance as the author of the envelope and acceptor of its proof under FD-BL-D1, and with the asset owner-operator** — unchanged from FD-BL §4.5 and FD-BL-D1 §8. This determination adds that the accountable human is entitled to a trustworthy basis: an authorization granted against output that does not satisfy I1–I3 is not a discharge of the FD-BL §4.5 obligation, and responsibility for supplying the independently verifiable basis and the corroboration path rests with the asset owner-operator. A supplier may build the indication and verification chain; a supplier may not hold the accountability, and no software component holds it.

9. Relationship to adjacent open items

- **Mode of discharge (FD-BL §6.1 / FD-BL-D1).** Fully coupled. FD-BL-D1's advance discharge is valid only on inputs satisfying I1–I3; this determination settles the input-integrity floor D1 §9 held open.
- **Terminology reservation (FD-BL §6.3 / FD-EV).** Resolved. This determination uses "indication" for output of an in-scope component functioning as the basis for an authorization, and "independently verifiable" for the anchoring path; it uses "envelope" only by reference to FD-BL-D1, in the authorization sense (Sense G) now reserved by **FD-EV**. Its usage is conforming and unchanged; the operational sense is renamed "operating band" by FD-EV. See FD-EV.

10. Conformance and amendment set

This determination governs the integrity of the basis on which an FD-BL §4.5 / FD-BL-D1 authorization is granted. Instruments carrying operative indication, decision-support, or gatekeeper-input requirements shall conform to it and shall **propagate by reference** (they cite FD-BL-D2; they do not restate it). The amendment set:

- **FD-BL-D1 §9** — the indication-integrity paragraph is superseded; FD-BL-D1 conformed at its v0.2, its §9 points to FD-BL-D2, and the interim-floor language is retired.
- **DBA-W §5.5.4** — the provision Pass 2 held and annotated closes here: its independent-verification requirement is the human-case instantiation of I3, and its "no actionable-default / no pre-populated input" rule is the operator confirmation step at §7. Remove the "HELD — pending §6.2" annotation and cite FD-BL-D2, I1–I3.
- **Gatekeeper-input provisions** — any provision where a gatekeeper's or classifier's *inputs* carry the safety argument (the indication-integrity dependency generally) is conformed to I1–I3 and cites FD-BL-D2 rather than the FD-BL-D1 §9 interim floor.
- **DBA-GAS-FC2** — its systematic-encroachment detection is harvested as an instantiation of the I3 corroboration path (divergence between the measured trajectory and normal behavior triggers human review); cite FD-BL-D2 alongside the existing FD-BL-D1 R2 citation.
- **Sector annexes (nuclear; energy; water)** — instantiate §7: the grade of independently verifiable, the corroboration path, denied-indication behavior, and the operator confirmation step.

The amendment set is small because Pass 2 already isolated and annotated the §6.2-coupled provisions rather than closing them; this determination lets them close.

11. Open at this version

- Quantitative standards for "independent of failure" between a generative synthesis and its confirming path — how much diversity or redundancy is enough for a given consequence class — are not standardized; this determination requires independence and corroboration but leaves the grade to the sector annexes.
- Methods for detecting slow, correlated drift that a coarse cross-check would miss (a synthesis and its nominal confirming channel drifting together) are left to the sector monitoring specification.
- The interaction of I3's corroboration requirement with FD-BL-D1 R2's completeness proof — whether corroboration is an input to the proof or a runtime check outside it, and which governs when a proof was accepted on inputs later found non-corroborable — is left to the Command Broker specification. It is open at this version and is the seventh open question of the independent review of 29 September 2026.

Revision History

0.5 — 15 October 2026. Issued in the November 2026 Edition, and recorded in FD-REG v0.32 in the same act. §4.3 gains a paragraph, *Sufficiency*, beside the requirements on traceable and corroborated inputs, carrying the text queued in FD-REG v0.30 from Determination_2026-10-14b_Enterprise_Network_Changes_Count_and_Indication_Integrity_Requires_Sufficiency: an indication can be correct and insufficient; what is missing is identified and sought before the authorization is relied on, and a basis acted on while incomplete is recorded as incomplete. The mechanism is open at this version. I1–I3 are unchanged and no section is renumbered.

0.4 — 2 October 2026. Issued in the October 2026 Edition, and recorded in FD-REG v0.22 in the same act. One cumulative entry for the step of the Action Plan for the October 2026 Edition that opens this instrument (Step 5).

Step 5, 2 October 2026 (F-04, F-10; F-12, F-13, F-22, F-23; Findings S2-2 and S4-1; F-25 checked). **F-04:** under Decision D, Option D1, this instrument stands and FD-BL moves; §2.3 now cites FD-BL §4.2 for the placement of output that makes no change of its own, rather than stating it. **F-10, on the author's decision of 2 October 2026:** §4's operative clause is restated so that output described in I1 carries the obligation, which is discharged only where I2 and I3 hold, and otherwise the output may not serve as a basis or as a placement or discharge input. I1 is widened to inputs on which placement or discharge depends, so that §5's statement that the classifier's inputs are bound now has a trigger in the rule. This is a small expansion of the obligation. **Finding S2-2:** I1, §2.2 and §9 speak of the output of an in-scope component (FD-LD §4.3) rather than generative output, and I1 adds a reading rule: the instrument's other references to a generative system, synthesis or recommendation name the principal case, and their requirements apply to any in-scope producer. **Finding S4-1:** §1 says advance discharge is by the Command Broker with a verified gatekeeper admitting. §8 names the Command Broker in both modes, following Decision C. **F-12:** §1 and §2.2 describe FD-BL §6.2 in the past frame. **F-13:** the header and §5 cite FD-BL §5 for the situational-awareness example, and §2.3 cites FD-BL §4.2. **F-23:** I1 to I3 are numbered §4.1 to §4.3, keeping their labels. **F-22:** the date line follows the working-master convention. **F-25:** MOC, APC and BES do not occur here. The review's seventh open question stays open in §11 and is named there. **On the author's further decision of 2 October 2026, I3 gains one sentence for classifier inputs:** where output subject to I1 serves as an input on which placement depends and cannot be corroborated through an independent path, the action it bears on is routed as above the line. This fails safe in the same way I3 does for an authorization, and states what I3 requires of an input that has no authorizer.

Verification read, 2 October 2026. §2.3 and §8 cite FD-BL §4.5 in full, because this instrument has no §4.5 of its own. No change to placement, granularity, scope or obligation.

Review of 2 October 2026. On the author's instruction to apply the cold review of the October 2026 Edition review draft as recommended (FD-BL Reconciliation/Independent_Review_2026-10-02_FD_October_2026_Edition_Review_Draft_v0_1), this instrument takes findings G-01, G-09, G-13; each edit is quoted in the review's application record.

0.3 — September 2026. Governing-instrument line conformed to the citation-form determination: FD-BL is named without a version token and the version is left to FD-REG. The previous line cited FD-BL v0.2 §6.2, a section that FD-BL v0.3 had closed to a pointer, so the citation directed a reader to the superseded text because that was the only text in which the cited section still stood. No change to I1–I3, the unified human/machine rule, or the amendment set.

0.2 — August 2026. Conformed the §9 terminology-reservation bullet to FD-EV: the envelope hold is discharged and the reserved-sense usage is confirmed conforming. No change to I1–I3, the unified human/machine rule, or the amendment set.

0.1 — August 2026. Initial issue, resolving FD-BL v0.2 §6.2 and settling the interim floor at FD-BL-D1 §9. Determines that generative output which can function as the basis for an above-line authorization carries an indication-integrity qualifier — attaching by function (I1), requiring traceability to independently verifiable values (I2), and requiring independent corroboration as a hard condition (I3) — without varying placement, granularity, or scope; unifies the human-broker (§4.5) and gatekeeper-input (FD-BL-D1) cases under one rule; forecloses both "below-line-therefore-unconstrained" and "feeds-authorization-therefore-above-line"; places the grade of "independently verifiable," the corroboration path, denied-indication behavior, and the operator confirmation step at the sector annex; and sets the amendment set that lets the Pass 2-held §6.2 provisions (DBA-W §5.5.4 first) close.